
The Song, Not the Pin: Route Transfer and Meaning-Based Place Identity in Collective Memory

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Abstract

Problem. The companion warp paper measures whether an agent can reach a target known only from a peer’s memory. Two assumptions remain hidden in that construction — and in essentially all shared-memory multi-agent systems. First, what transfers is a *point*: broadcast snapshots carry place summaries but no edges, so the traveler learns WHERE and must re-derive HOW. Second, place identity is a *shared coordinate frame*: ‘the same place’ means ‘close (x, y) in a grid the environment hands to every agent’; semantics only breaks ties inside a spatial radius. In the source culture a songline is neither: the song IS the route, and its places are identified by what they look like, not by coordinates nobody shares. **Approach.** We remove both assumptions and measure what each was hiding. (i) *Route warp*: broadcasts carry traversed edges with provenance; φ_{route} is the foreign share of edge mass along the committed path, and a strict RW^* is a connected foreign sub-path the traveler has never walked. (ii) *Semantic place identity*: places are matched by local semantic constellations (translation-invariant fingerprints); the inter-agent frame — translation in one step, the discrete subgroup $\mathbb{Z}^2 \rtimes C_4$ of $SE(2)$ (translations and quarter-turn rotations) in the next — is recovered from mutually-unique matches of co-visited landmarks, and foreign evidence, including never-seen targets, is transported through it. **Findings.** What route knowledge buys is a step function, not a slope: place-only transfer collapses at the FIRST wall (success 1.0 at detour $D=1.00$ vs. 0.0 at $D=1.38$, and worse than blind exploration), while route transfer completes everywhere. Aging breaks the transferred route at a point predictable to the exact edge (110/110 regimes; all 40 mid-route ruptures across two world classes at the predicted index, error 0), and safety travels with the route: 0 hazard hits over the witness’s path vs. 3.25 [2.45, 4.10] for the place-only traveler, with the risk RESUMING at the predicted rupture point. Meaning-based identity recovers secret frames exactly (100% translation; 80/80 under $SE(2)$), restores 88% of the shared-frame performance in the full peer stack, and fails *closed* under perceptual aliasing where the coordinate assumption fails *open* — and the companion distance law re-emerges tick-exactly (breakpoints 20/14/9) with no shared origin and no shared north. An ablation of the landmark vocabulary shows the identity machinery is *fail-safe in its feature alphabet*: impoverishing the vocabulary down to a single tag class costs coverage (exact recovery $1.00 \rightarrow 0.83$), never correctness (zero fail-open transports in 252 cells). Finally, a song-anatomy experiment separates the roles inside the song itself: a purely relational song (landmark signatures + beat vectors, *no global coordinates at all*) transports a never-seen target exactly at 6.4% of the snapshot bit cost, while beats alone fail *open* (11/11 wrong locks under anchor disagreement) and signatures alone cannot transport the unseen at all. **Claim.** We claim that the transferable unit of navigation memory is not a destination coordinate or an isolated place record but a provenance-bearing relational route: landmarks establish identity, edges provide

transport, and their composition lets an agent execute another agent’s experience without sharing a coordinate frame. **Thesis.** A route description without identity is unsafe, and a place identity without transport is inert; executable memory requires their composition. A shared map is not what makes collective navigation work — what transfers is the song, and each impoverishment of it (point instead of route, coordinates instead of meaning) fails in a specific, measurable, and now predicted way. **Scope.** Grid worlds; translation and 90° rotations; measurement and minimal mechanisms, not a general theory.

1 Introduction

A songline encodes a journey: a sequence of places, each recognisable by what is there, bound into a path by the song’s order. When multi-agent systems “share memory,” they typically share something much poorer — a pin on a common map: coordinates of a target in a frame all agents are assumed to inhabit. The companion warp paper¹ made the act of navigating by someone else’s memory a measured event (W^* : a materialisation whose evidence is entirely foreign) and showed its value is governed by the coordination contract attached to the information. But its construction inherited the two impoverishments silently:

1. **The pin, not the song.** Broadcast snapshots carry place concepts — centroids, tags, confidences — but no edges. The receiver learns WHERE the water is; the route is re-derived from scratch. In open grids this is invisible: the manhattan path is the route. The moment paths are expensive, it is everything.
2. **Coordinates, not meaning.** ‘Same place’ conflates two questions the series never separated. The *semantic predicate* — is this the same KIND of desired object (water, GOOD)? — travels in every snapshot and is what task satisfaction is defined over. *Entity correspondence* — is this the same world referent? — is operationalised as coordinate proximity, with semantics as a tie-breaker inside a radius. The cross-agent place-correspondence problem — the reason map merging is hard in robotics — is bracketed by fiat.

This paper removes both assumptions, one at a time, with the measurement discipline of the series: every claim is registered before the runs, failures against registration are reported with their mechanisms, and the strongest results are exact-match validations of closed-form predictions.

What we claim. We claim that transferring only WHERE a target is located is insufficient whenever success depends on HOW it was reached, and that shared coordinates are unnecessary when places re-identify through mutually-unique semantic constellations. The paper supports five empirical claims. **(EC1)** The benefit of route over place-only transfer is a cliff: place-only transfer collapses at the first obstructing wall and can be worse than blind exploration, while route transfer stays executable across detour regimes. **(EC2)** The rupture law predicts the exact edge at which a transferred route loses validity — all 110 regimes, all 21 mid-route ruptures, zero edge-index error. **(EC3)** Route transfer preserves the witness’s measured risk profile only over the validated route prefix; after rupture the risk returns at the predicted edge. **(EC4)** Meaning-based landmark correspondence recovers private frames under translations and quarter-turn rotations, while ambiguous cases degrade through refusal rather than incorrect materialisation. **(EC5)** Landmarks and edges implement distinct functions — signatures without edges cannot reach an unseen target, edges without a valid anchor fail open — and their composition matches full-snapshot transport at 6.4% of the encoded bit cost.

Contributions.

1. **Route warp** ($RW^*, \varphi_{\text{route}}$): edge-level provenance and the strict route-warp event — a connected foreign sub-path the traveler never walked (§3). The value of route over place transfer is a *cliff*, not a slope: place-only transfer dies at the first wall ($1.0 \rightarrow 0.0$ success between measured detour 1.00 and 1.38) and is **WORSE** than no transfer at all there, while route transfer completes everywhere (§3.1).

¹Companion submissions: the Q/R/M/C measurement framework, the collective-semantic-memory architecture study, and the semantic-warp measurement paper. Citations deferred per concurrent-submission policy.

- 89 2. **The rupture law:** a witness stamps edges sequentially, so its trace ages non-uniformly and a
 90 delayed traveler chases a fading song. Replaying the series' trust $\times$ staleness gate over calibrated
 91 kinematics predicts *where the song breaks* to the exact edge: 110/110 regime matches, all 21
 92 mid-route ruptures at the predicted index (error 0), and a discrete trust-flip that zeroes route
 93 commits entirely (§4).
- 94 3. **Hazard stratification:** the wall and hazard axes are independent — walls break place-transfer's
 95 *completion*, hazards break its *safety* (0 vs. 3.25 hits at equal completion and +0.7 ticks), and
 96 when the song ruptures the risk resumes at the predicted edge (§5).
- 97 4. **Meaning-based place identity:** translation-invariant semantic fingerprints, mutually-unique
 98 landmark matching, and consensus frame recovery — translation first, full $SE(2)$ with secret
 99 90° rotations after (§6). Exact frame recovery (100%; 80/80 under $SE(2)$); online alignment
 100 in the full peer stack recovering 88% of the shared-frame gap; a proved failure asymmetry
 101 (semantic identity fails *closed* under aliasing and under in-principle-unrecoverable 180° sym-
 102 metry; coordinate identity fails *open*); and the companion distance law reproduced tick-exactly
 103 (20/14/9) with no shared frame at all (§6.1). A registered landmark-vocabulary ablation shows
 104 the machinery is *fail-safe in its feature alphabet*: down to a single content class, impoverishment
 105 converts recovery into refusal — never into a wrong lock (0 fail-open transports in 252 cells;
 106 §6.2). A song-anatomy experiment then separates the roles within the song: *nodes carry identity*,
 107 *edges carry transport* — beats alone fail open, signatures alone cannot reach the unseen, and
 108 their composition matches full-snapshot transport at 6.4% of the bits (§6.3).

109 2 Setting

110 Grid worlds (12×10 to 14×12) with semantic water targets, hazards, and walls; agents with symbolic
 111 place memory, peer broadcast on cadence K , and the series' standard planner (target lock = the
 112 Q/R/M/C M^* event; completion C^*). The trust $\times$ staleness inclusion rule of the collective-memory
 113 companion is used throughout with unchanged constants: evidence mass trust $\cdot e^{-\alpha \text{age}} \cdot m$, threshold
 114 $\tau=0.30$, $\alpha=0.05$. Warp events follow the companion warp paper: provenance φ frozen at lock time;
 115 completion attributed to a continuous lock from a foreign commit ($\varphi=1$ at first lock) to success.

116 3 Route warp: transferring the HOW

117 **Edge provenance.** Broadcast snapshots additionally carry the sender's OWN traversed edges
 118 ($a, b, \text{count}, \text{last tick}$); foreign edges are never re-broadcast (the same privacy contract as place
 119 snapshots: no agent reads another's graph, transport only). For the path P a planner commits,

$$\varphi_{\text{route}}(P) = \frac{\sum_{e \in P, \pi(e) \neq \text{self}} w(e)}{\sum_{e \in P} w(e)}, \quad w(e) = \text{trust} \cdot e^{-\alpha \text{age}(e)} \cdot \log(1 + \text{count}(e)),$$

120 the same weights the route gate uses. A **strict** RW^* is a committed, traversed, connected sub-path of
 121 ≥ 5 edges with $\varphi_{\text{route}} = 1$: the whole verse is someone else's. Place warp is the degenerate case (path
 122 length 0: only the target is foreign), giving a two-dimensional provenance plane ($\varphi_{\text{place}}, \varphi_{\text{route}}$).

123 **First strict RW^* .** On a comb maze with measured detour $D=2.65$, a traveler commits and walks
 124 a fully foreign 45-edge route to completion in 54 ticks; with the same foreign PLACE evidence but
 125 no edges, the standard planner never arrives (300-tick budget), and blind exploration takes 126.

126 3.1 The cliff: what route knowledge buys

127 Figure 1 shows both faces of the result. Two pre-registered predictions failed here, in instructive
 128 ways. The negative control at $D=1$ was registered in ticks ($|\text{gain}| \leq 3$) and came out +4.7: the gap
 129 is *turn economy* (BFS routes walk straight legs; the dominant-axis planner alternates axes), not route
 130 knowledge — in path LENGTH the gap is exactly 0.0 [0.0, 0.0] across all 20 open-grid cells. And
 131 “gain grows with D ” is unmeasurable: with place success at 0 the gain saturates at the censoring
 132 ceiling. The correct form of the law is a step located by the probe bucket at $D \in (1.00, 1.38)$, i.e. the
 133 first wall.

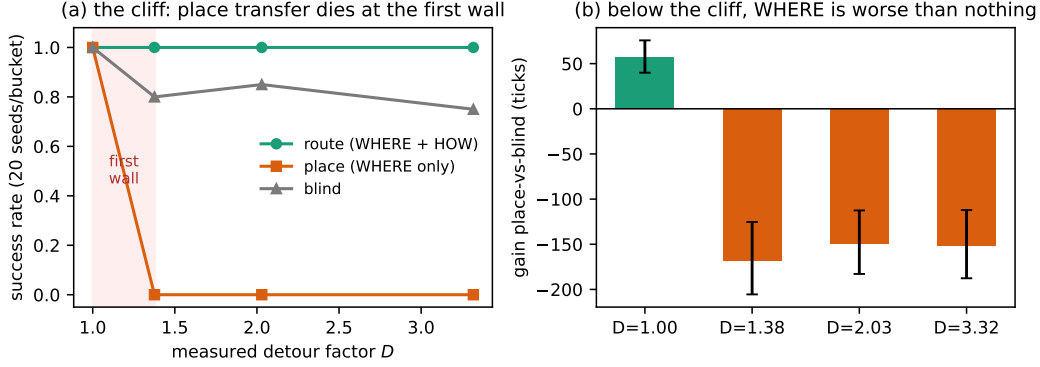


Figure 1: The cliff (paired arms on identical mazes, 20 seeds per measured-detour bucket). **(a)** Place-only transfer does not degrade with detour — it collapses at the FIRST wall, located by the probe bucket between $D=1.00$ and $D=1.38$; route transfer completes everywhere. **(b)** Below the cliff, knowing WHERE without HOW is *worse than knowing nothing*: the known-target manhattan pull pins the planner against walls (−150 to −168 ticks vs. blind exploration), while in the open world the same knowledge is worth +57.

Table 1: Three arms on identical mazes (paired by seed), bucketed by MEASURED detour factor D ; 20 seeds per bucket, censored gains (fail = 300). Place-only transfer does not degrade with D — it collapses at the first wall, and is worse than blind exploration there (the known-target manhattan pull pins the planner against walls). Route transfer completes everywhere; every completion is a strict RW^* with $\varphi_{\text{route}} = 1$.

D	gain route-vs-place [CI]	gain route-vs-blind [CI]	succ r/p/b
1.00	+4.7 [+3.4, +6.0]	+62 [+44, +80]	20/20/20
1.38	+281 [+279, +283]	+113 [+77, +155]	20/0/16
2.03	+273 [+271, +275]	+123 [+91, +160]	20/0/17
3.32	+262 [+259, +264]	+110 [+72, +152]	20/0/15

134 4 The rupture law: where the song breaks

135 The witness stamps edge i at tick i : its trace ages non-uniformly, the start of the route always
 136 older than the end, and a traveler delayed by a_0 *chases a fading song*. Under the standard gate the
 137 binding edge is the CURRENT one (the oldest remaining), giving three regimes on the a_0 axis:
 138 complete; rupture at a predictable path index; dead on arrival. The predictor is fully deterministic
 139 — a no-decay calibration run records the traveler’s kinematics (path index per tick, turns included),
 140 and the registered prediction replays the closed-form gate over that trajectory, the same device that
 141 produced the companion paper’s 6/6 hold-out.

142 Across 5 mazes $\times$ 11 delays $\times$ 2 trusts (110 cells, registered before execution): **110/110 regime**
 143 **matches; all 21 mid-route ruptures at exactly the predicted edge index (error 0)**; and a route
 144 trust-flip — at trust 0.7 the single-traversal edge mass gives $\text{age}_{\text{max}} = 9.6 < L$, so *no route ever*
 145 *commits* (0 commits observed; the discrete off-switch of the companion law, now for paths). The
 146 rupture of a transferred route is thus not noise but a law: the same constants (α, τ , trust) that set
 147 WHEN a place-warp gate closes set WHERE a route-warp breaks.

148 5 Hazard stratification: the song protects

149 The traveler following the witness’s path preserves the witness’s measured risk profile within the
 150 validated route prefix, at a time premium of +0.7 ticks (Figure 2a). Composing with §4: in the
 151 rupture regime (19/20 worlds) the traveler takes **0 hits before the rupture** and 0.58 [0.11, 1.32] after
 152 it, when it falls back on place knowledge and re-enters the field — with all 19 ruptures again at
 153 the R2-predicted index (error 0; the edge-exact predictor replicated on a new world class with no

Table 2: Open fields dense with passable-but-punished hazards (density 0.20), the witness’s route hazard-avoiding by experience; 20 paired worlds, decay off (the safety axis isolated from the aging axis). Both transfer modes COMPLETE — the wall axis (Table 1) and the hazard axis are independent; they differ in the price.

Arm	hazard hits [CI]	completion	mean t
route (foreign safe path)	0 (all 20 cells)	20/20	16.6
place (WHERE only)	3.25 [2.45, 4.10]	20/20	15.9
blind	28.7	—	—

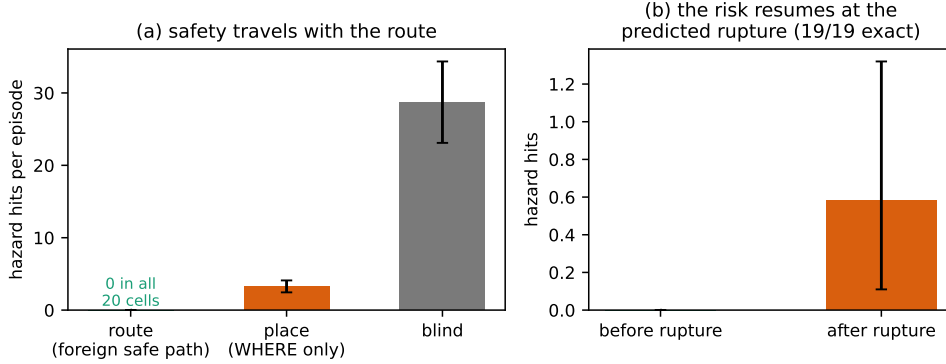


Figure 2: The song protects. **(a)** Hazard hits per episode: the traveler on the witness’s route preserves the witness’s measured risk profile within the validated route prefix (0 in all 20 worlds); the place-only traveler pays 3.25 hits for the same completion; blind exploration pays 28.7. **(b)** Composition with the rupture law: in the rupture regime the traveler takes zero hits while the song lasts, and the risk resumes after the break — which occurs at the R2-predicted edge index in all 19/19 cells.

154 adjustment). The song protects while it lasts, and it stops protecting exactly where the law says the
 155 verse ends.

156 6 Meaning-based place identity

157 Everything so far — and everything in the companion papers — rides on a shared coordinate frame.
 158 We now remove it: each agent’s private frame is the true grid under a secret translation (later: plus a
 159 secret 90°-multiple rotation), and broadcasts carry private coordinates that are meaningless to the
 160 receiver until it decides what the sender’s places MEAN.

161 **Two levels of ‘the same’, and what a landmark is.** Cross-agent place identity has two levels
 162 that must not be conflated. The *semantic predicate* says two cells belong to the same desired class
 163 (*water_source*, *GOOD*): it travels in every snapshot, and it is what task satisfaction is defined
 164 over. But no predicate can decide *entity correspondence* — that two memory entries denote the
 165 same world referent — because a predicate is exactly what aliases: all water looks like water. Route-
 166 relative description (‘ n steps past the last *GOOD*’) does not decide it either: it is provenance of a
 167 path, not identity of its endpoints (§3–§4 measure what it buys and where it breaks). What decides
 168 correspondence in this series is a *landmark*: a place whose local semantic constellation supports a
 169 mutually-unique match across agents. Landmark is a *role*, not a cell type — which tag classes can
 170 play it is itself measured below (§6.2).

171 **Fingerprints and the handshake.** A place fingerprint is the local semantic constellation around a
 172 visited cell — which content tags (hazards, walls, water) sit at which relative offsets — translation-
 173 invariant and coordinate-free. Matching is *mutually unique* (a pair is accepted only if each side is the
 174 other’s sole candidate above an exact-equality threshold; the sender knows far more places than the
 175 receiver, and one-directional uniqueness aliases distant look-alikes into the receiver’s neighbourhood),
 176 fingerprints need ≥ 2 salient keys, and the frame translation is the modal delta of ≥ 3 agreeing pairs.

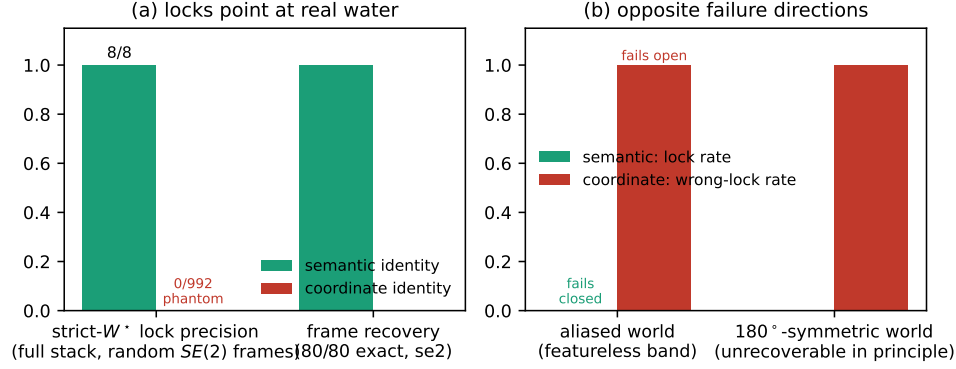


Figure 3: Meaning-based identity under $SE(2)$ frames. **(a)** Every strict- W^* lock the semantic stack makes points at real water (8/8); the coordinate stack makes 992 locks, all phantoms (0/992); frame recovery itself is exact in 80/80 witness-traveler combinations. **(b)** The failure asymmetry in worlds where identification is impossible (featureless band; a 180° -symmetric layout where the frame is unrecoverable in principle): semantic identity refuses to lock — fails closed — while coordinate identity keeps confidently locking wrong cells — fails open.

Landmarks must carry at least one CONTENT tag: border-only ('void') patterns describe the shape of the world, not a place, and are excluded — the lesson of the corner trap below.

Witness-traveler results. Over $4 \text{ offsets} \times 2 \text{ hazard densities} \times 10 \text{ seeds}$: the semantic aligner recovers the secret offset *exactly* in 100% of landmark-world runs (mean 15.1 mutual pairs) and completes with strict W^* ($\varphi=1$) every time; the coordinate baseline under any nonzero offset locks a non-water cell in 100% of runs (its rare 'successes', 6.7%, are the displaced route accidentally crossing the true water). In the aliased world (featureless interior band) the two identity regimes break in OPPOSITE directions: semantic identity makes *zero* locks — it fails closed — while coordinate identity keeps confidently locking wrong cells — it fails open. This asymmetry, not raw accuracy, is the deepest argument for meaning-based identity: when the world does not support identification, the semantic system knows it.

6.1 The full stack, and removing the last axis

Online alignment in the peer stack. With $N=4$ agents under scarcity, all in distinct secret frames, alignment develops DURING the episode: agents explore, co-visited landmarks accumulate, and per-sender frames lock in mid-run (onset in 100% of episodes, mean tick ≈ 15). Pooled success: shared-frame oracle 0.456, coordinate under misalignment 0.350, semantic 0.444 — **88% of the gap recovered**, with strict W^* functioning through recovered frames.

The law without the frame. Re-running the companion distance law with the trust $\times$ staleness gate ON TOP of semantic identity under a frame offset gives breakpoints 20/14/9 — *the same integers as the shared-frame closed form, exactly*. The gate and the frame recovery are orthogonal layers: the law does not notice that the shared grid is gone.

$SE(2)$: no shared north. With secret rotations added, one rotation hypothesis per 90° reuses the translation aligner and the winner must *dominate* ($\geq 2 \times$ the runner-up's consensus). Results: exact (r^*, δ^*) recovery in **80/80** rotation $\times$ offset $\times$ seed combinations with 100% strict- W^* completion; in a 180° -symmetric world — where the frame is unrecoverable *in principle* — the aligner refuses (0/10 locks) while coordinates fail open (10/10 wrong locks); and the distance law again does not notice (bp = 20 under a 90° -rotated witness). The translation-only aligner, for contrast, never recovers a rotated frame (0/60) but does not always fail closed: in 2/60 cells a 180° -symmetric hazard substructure lets it fail OPEN — the empirical motivation for the dominance rule.

The corner trap. In the full stack, raw success is non-discriminating under rotations (rotated coordinate poison mostly maps off-grid and degrades into pseudo-exploration), so the registered

discriminator is *warp-lock precision*: the share of strict W^* locks pointing at real water. Before the void filter, early border fingerprints (agents start in corners; rectangle borders are 180° -symmetric) fed wrong-frame hypotheses and semantic precision was 0.159; with borders excluded as non-landmarks it is 1.000 (8/8) **vs.** 0.000 (0/992 **phantom locks**) **for coordinates**, with alignment onset in 100% of episodes and clean regressions on all earlier experiments (Figure 3).

6.2 What makes a landmark: the vocabulary ablation

The identity machinery rides on a fixed landmark vocabulary $\Sigma = \{\text{wall, hazard, water, void}\}$: the tag classes allowed into a fingerprint. Every claim above is therefore conditional on a hand-chosen alphabet. The last experiment makes the alphabet the treatment: witness-traveler frame recovery (worlds carrying both scattered walls and scattered hazards; 3 offsets $\times$ 12 seeds per arm) under seven vocabularies, from full Σ down to single classes.

Table 3: Landmark-vocabulary ablation (36 cells per arm, 252 total; registered predictions A1–A4, all PASS). Impoverishing the alphabet costs *coverage* — recovery converts into refusal — but never *correctness*: every transported lock, under every vocabulary, goes through the exactly-recovered frame. Attribution = share of consensus-pair key mass under full Σ .

Vocabulary	exact recovery	mean pairs	lock precision	fail-open
full Σ	1.00	33.2	1.00	0
– void	1.00	32.0	1.00	0
– hazards	0.92	11.8	1.00	0
– walls	1.00	15.5	1.00	0
hazards only	1.00	13.7	1.00	0
walls only	0.83	10.8	1.00	0
void only	0.00	0.0	—	0

Four registered predictions, four passes (Table 3). **(A1) The ablation is fail-safe:** in 252 cells there is not one fail-open transport — the fail-closed asymmetry of §6 holds not just against ambiguous *worlds* but against impoverished *alphabets*. **(A2) Void is scaffolding, not identity:** border patterns carry 0.000 of the consensus key mass, and dropping them changes nothing — consistent with the corner trap, where they actively poisoned rotation hypotheses. **(A3) One content class suffices:** hazards alone recover every frame; walls alone recover 0.83 and refuse the rest. Correspondence needs only ≥ 3 mutually-unique constellations over *some* content class — a bounded-rationality budget, not a rich ontology. **(A4) Attribution is measurable:** hazard constellations carry 0.56 of the consensus mass, walls 0.44. Which features function as landmarks is thus not a modelling assumption that happened to work — it is a measurable property of the world $\times$ vocabulary pair, and the machinery degrades along the safe axis whenever the vocabulary is wrong.

6.3 Song anatomy: nodes carry identity, edges carry transport

§6 opened with a claim we have so far only argued: route-relative description (*‘n steps past the last GOOD’*) is provenance of a path, not identity of its endpoints. The last experiment measures it. A witness compresses its route to the water into a purely *relational* song — a sequence of couplets (σ, e) , where σ is the landmark constellation signature and e the exact beat vector from the previous couplet — carrying *no global coordinates whatsoever*. A band-limited traveler who has never seen the water receives it. Four registered arms (bit codec fixed before the runs; 12 paired worlds):

- **Beats only** (the pure *‘n steps from the last GOOD’*): usable only by *assuming* the anchor. With a shared anchor it is exact (11/11, the negative control); with anchor disagreement it locks every time and is right *never* (11/11 fail-open mislocks, displaced by exactly the anchor offset). Beats do not carry identity — they parasitise on the identity of the anchor.
- **Signatures only:** re-identifies co-visited places but has no edges to walk — *zero* transports of the never-seen target. Nodes alone cannot reach the unseen.
- **Signatures + beats:** couplets matched mutually-uniquely, the beat chain checked for metric consistency between matched couplets (a loop-closure test), the target dead-reckoned from the last matched landmark. Coverage 1.0, every transport exact, zero fail-open — at 366 bits per song.

247 • **Full snapshots** (the machinery of §6): the information upper bound — same correctness at 5,690
248 bits.

249 All four registered predictions passed. The song is not a lighter snapshot; it is a different *kind* of object:
250 identity lives in the nodes (landmarks), transport lives in the edges (beats), and their composition
251 delivers full-snapshot transport at 6.4% of the bit cost while inheriting the fail-safe property of §6.2
252 — everything that goes wrong goes wrong as a refusal. (This 6.4% is the *pure song codec*: nodes and
253 edges only. A deployed collective-memory protocol adds provenance, timestamps, world versions,
254 and a utility certificate; the full-protocol payload ratio, measured in the companion formation paper,
255 is correspondingly larger.)

256 7 Related work

257 **Map merging and place recognition.** Multi-robot SLAM treats inter-map data association as
258 an estimation problem over metric features Cadena et al. (2016); visual place recognition Lowry
259 et al. (2016) matches appearance. Our contribution is not a competing matcher but the measurement
260 layer on top: provenance-conditioned completion through recovered frames, the fail-closed/fail-open
261 asymmetry as a registered prediction, and law invariance under frame removal.

262 **Route sharing.** Stigmergy transfers paths anonymously (pheromone gradients); learned commu-
263 nication (e.g. CommNet Sukhbaatar et al. (2016)) transfers policies opaquely. Neither exposes an
264 edge-provenance decomposition, so neither can state — let alone predict — where a transferred route
265 breaks. The rupture predictor is Time Warp Jefferson (1985) thinking applied to memory decay:
266 deterministic replay of a gate over calibrated kinematics.

267 **Songlines.** The narrative frame is now load-bearing in both directions: the song is a sequence (route
268 warp), its places are recognised by meaning (semantic identity), it fades verse by verse (rupture law),
269 and it protects the walker while it lasts (hazard stratification).

270 8 Limitations and honest negatives

271 (1) Grid worlds; translation and 90° rotations only — continuous $SE(2)$ and appearance noise are
272 future work (the fingerprint threshold is exact-equality because the substrate is noise-free; noisy
273 substrates need the margin machinery of place recognition). (2) Four registered predictions failed
274 and are reported with mechanisms: the $D=1$ negative control in ticks (turn economy; exact in path
275 length), gain monotonicity in D (censoring; the truth is a cliff), the full-stack success comparison
276 under rotations (non-discriminating; replaced by lock precision), and “translation-fails-closed” (2/60
277 fail open via symmetric substructures). (3) The route follower is a new planner tier; all three arms
278 share it, but route-warp value is measured relative to this consumer. (4) Hazard safety transfers by
279 construction when the witness’s route is safe; the measured content is that place transfer CANNOT
280 carry it and that the protection ends at the predicted rupture. (5) Scarcity-level completion of warp
281 locks remains governed by the collision phenomenology of the companion paper. (6) The landmark
282 vocabulary Σ is hand-fixed; §6.2 measures its ablation but no mechanism *selects* it. What earns a
283 feature landmark status here is our choice of alphabet plus the world’s cooperation; a mechanism
284 in which features earn that status by participating in successful alignments — the vocabulary itself
285 shaped by coordination, not designed — is the natural next step of the series and is not attempted
286 here.

287 9 Conclusion

288 Strip the song down to a pin on a shared map and two things break invisibly: the HOW (fatal
289 at the first wall, expensive in every hazard field) and the WHO-MEANS-WHAT (fatal under any
290 frame disagreement, and dangerous precisely because it fails open). Restore the song — edges
291 with provenance, places with meaning — and both failures become measurable, lawful, and largely
292 repaired: routes transfer completion and safety and break only where the aging law says; meaning
293 recovers frames exactly, refuses honestly when the world is ambiguous, and carries the entire
294 measurement stack — φ , gates, distance law — unchanged into a world where agents share no

coordinates at all. Even the alphabet the song is written in turned out to be measurable rather than sacred: identity survives vocabulary impoverishment fail-safe, losing coverage but never truth. The pin was never the memory. The song is — and the open question the ablation sharpens is not which features ARE landmarks, but which features a collective’s own coordination will MAKE into landmarks.

300 **A Registered predictions and revisions**

301 All predictions were written to disk before execution; the artifact retains both versions wherever an
 302 operationalisation was revised. Revisions: R1 negative control re-measured in path length; R1 gain
 303 monotonicity re-formulated as cliff detection; W9 full-stack discriminator changed from success to
 304 warp-lock precision; W8 strict- W^* functioning extended to include warp-assisted own completions
 305 (taxi chains). The vocabulary ablation (A1–A4, §6.2) and the song-anatomy experiment (S0.1–S0.4,
 306 §6.3, bit codec registered with the predictions) passed as registered, with no revisions.

307 **B Reproducibility**

```
308 PYTHONPATH=. python experiments/warp/exp_route_warp_r0.py
309 PYTHONPATH=. python experiments/warp/exp_route_warp_r1.py --seeds 20
310 PYTHONPATH=. python experiments/warp/exp_route_warp_r2.py --seeds 5
311 PYTHONPATH=. python experiments/warp/exp_route_warp_r3.py --seeds 20
312 PYTHONPATH=. python experiments/warp/exp_warp_semantic_identity.py
313 PYTHONPATH=. python experiments/warp/exp_warp_semantic_stack.py --seeds 20
314 PYTHONPATH=. python experiments/warp/exp_warp_rotation.py --seeds 10
315 PYTHONPATH=. python experiments/warp/exp_warp_landmark_ablation.py
316 PYTHONPATH=. python experiments/song_grammar/exp_s0_song_smoke.py
```

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